

JEE Mains 2019 Chapter wise Question Bank

EMI - Questions

Q1

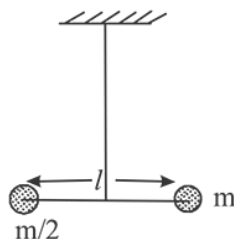
A conducting circular loop made of a thin wire, has area $3.5 \times 10^{-3} \text{ m}^2$ and resistance 10Ω . It is placed perpendicular to a time dependent magnetic field $B(t) = (0.4 \text{ T}) \sin(50\pi t)$. The net charge flowing through the loop during $t = 0 \text{ s}$ and $t = 10 \text{ ms}$ is close to:

- (1) 14 mC (2) 7 mC
(3) 21 mC (4) 6 mC

9 Jan Morning

Q2

Two masses m and $\frac{m}{2}$ are connected at the two ends of a massless rigid rod of length l . The rod is suspended by a thin wire of torsional constant k at the centre of mass of the rod-mass system (see figure). Because of torsional constant k , the restoring torque is $\tau = k\theta$ for angular displacement θ . If the rod is rotated by θ_0 and released, the tension in it when it passes through its mean position will be:



- (1) $\frac{3k\theta_0^2}{l}$ (2) $\frac{2k\theta_0^2}{l}$
(3) $\frac{k\theta_0^2}{l}$ (4) $\frac{k\theta_0^2}{2l}$

9 Jan Morning

Q3

A power transmission line feeds input power at 2300 V to a step down transformer with its primary windings having 4000 turns. The output power is delivered at 230 V by the transformer. If the current in the primary of the transformer is 5 A and its efficiency is 90% , the output current would be:

- (1) 50 A (2) 45 A
(3) 35 A (4) 25 A

9 Jan Evening

Q4

The self induced emf of a coil is 25 volts . When the current in it is changed at uniform rate from 10 A to 25 A in 1 s , the change in the energy of the inductance is:

- (1) 740 J (2) 437.5 J
(3) 540 J (4) 637.5 J

10 Jan Evening

Q5

There are two long co-axial solenoids of same length l . The inner and outer coils have radii r_1 and r_2 and number of turns per unit length n_1 and n_2 , respectively. The ratio of mutual inductance to the self-inductance of the inner-coil is :

- (1) $\frac{n_1}{n_2}$ (2) $\frac{n_2}{n_1} \cdot \frac{r_1}{r_2}$
(3) $\frac{n_2}{n_1} \cdot \frac{r_2^2}{r_1^2}$ (4) $\frac{n_2}{n_1}$

11 Jan Morning

Q6

EMI

A copper wire is wound on a wooden frame, whose shape is that of an equilateral triangle. If the linear dimension of each side of the frame is increased by a factor of 3, keeping the number of turns of the coil per unit length of the frame the same, then the self inductance of the coil:

- (1) decreases by a factor of 9
- (2) increases by a factor of 27
- (3) increases by a factor of 3
- (4) decreases by a factor of $9\sqrt{3}$

11 Jan Evening

Q7

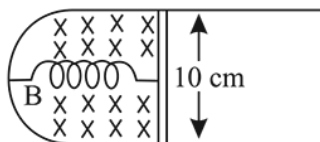
A 10 m long horizontal wire extends from North East to South West. It is falling with a speed of 5.0 ms^{-1} , at right angles to the horizontal component of the earth's magnetic field, of $0.3 \times 10^{-4} \text{ Wb/m}^2$. The value of the induced emf in wire is :

- (1) $1.5 \times 10^{-3} \text{ V}$
- (2) $1.1 \times 10^{-3} \text{ V}$
- (3) $2.5 \times 10^{-3} \text{ V}$
- (4) $0.3 \times 10^{-3} \text{ V}$

12 Jan Evening

Q8

A thin strip 10 cm long is on a U shaped wire of negligible resistance and it is connected to a spring of spring constant 0.5 Nm^{-1} (see figure). The assembly is kept in a uniform magnetic field of 0.1 T. If the strip is pulled from its equilibrium position and released, the number of oscillations it performs before its amplitude decreases by a factor of e is N. If the mass of strip is 50 grams, its resistance 10Ω and air drag negligible, N will be close to :



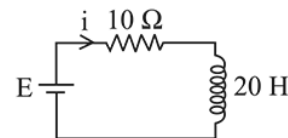
- (1) 1000
- (2) 50000
- (3) 5000
- (4) 10000

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Q9

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A 20 Henry inductor coil is connected to a 10 ohm resistance in series as shown in figure. The time at which rate of dissipation of energy (Joule's heat) across resistance is equal to the rate at which magnetic energy is stored in the inductor, is :



- (1) $\frac{2}{\ln 2}$
- (2) $\frac{1}{2} \ln 2$
- (3) $2 \ln 2$
- (4) $\ln 2$

8 April Morning

Q10

The total number of turns and cross-section area in a solenoid is fixed. However, its length L is varied by adjusting the separation between windings. The inductance of solenoid will be proportional to:

- (1) L
- (2) L^2
- (3) $1/L^2$
- (4) $1/L$

9 April Morning

Q11

The magnetic field of a plane electromagnetic wave is given by:

$$\vec{B} = B_0 \hat{i} [\cos(kz - \omega t)] + B_1 \hat{j} \cos(kz + \omega t)$$

Where $B_0 = 3 \times 10^{-5} \text{ T}$ and $B_1 = 2 \times 10^{-6} \text{ T}$.

The rms value of the force experienced by a stationary charge $Q = 10^{-4} \text{ C}$ at $z = 0$ is closest to:

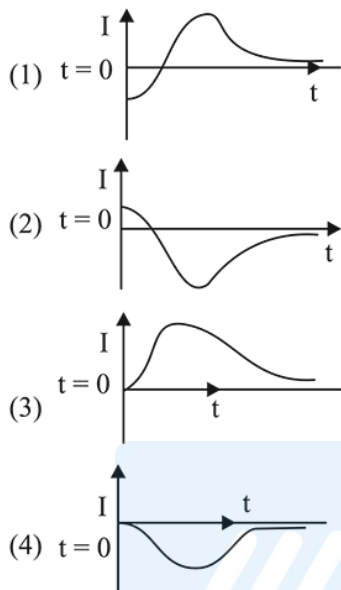
- (1) 0.6 N
- (2) 0.1 N
- (3) 0.9 N
- (4) $3 \times 10^{-2} \text{ N}$

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Q11

EMI

A very long solenoid of radius R is carrying current $I(t) = kte^{-\alpha t}$ ($k > 0$), as a function of time ($t \geq 0$). Counter clockwise current is taken to be positive. A circular conducting coil of radius $2R$ is placed in the equatorial plane of the solenoid and concentric with the solenoid. The current induced in the outer coil is correctly depicted, as a function of time, by:



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Q12

Two coils 'P' and 'Q' are separated by some distance. When a current of 3A flows through coil 'P', a magnetic flux of 10^{-3} Wb passes through 'Q'. No current is passed through 'Q'. When no current passes through 'P' and a current of 2A passes through 'Q', the flux through 'P' is:

- (1) 6.67×10^{-4} Wb (2) 3.67×10^{-3} Wb
(3) 6.67×10^{-3} Wb (4) 3.67×10^{-4} Wb

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Q13

A transformer consisting of 300 turns in the primary and 150 turns in the secondary gives output power of 2.2kW. If the current in the secondary coil is 10 A, then the input voltage and current in the primary coil are :

- (1) 220 V and 20 A (2) 440 V and 20 A
(3) 440 V and 5 A (4) 220 V and 10 A

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Q14

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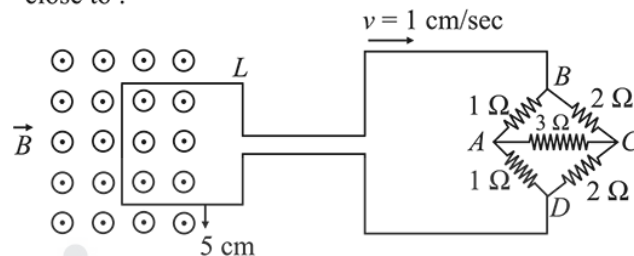
A coil of self inductance 10 mH and resistance 0.1Ω is connected through a switch to a battery of internal resistance 0.9Ω . After the switch is closed, the time taken for the current to attain 80% of the saturation value is [take $\ln 5 = 1.6$]

- (1) 0.324 s (2) 0.103 s (3) 0.002 s (4) 0.016 s

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Q14

The figure shows a square loop L of side 5 cm which is connected to a network of resistances. The whole setup is moving towards right with a constant speed of 1 cm s^{-1} . At some instant, a part of L is in a uniform magnetic field of 1 T, perpendicular to the plane of the loop. If the resistance of L is 1.7Ω , the current in the loop at that instant will be close to :

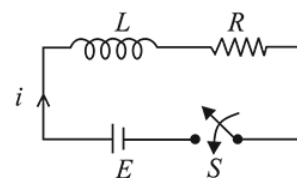


- (1) $60 \mu\text{A}$ (2) $170 \mu\text{A}$
(3) $150 \mu\text{A}$ (4) $115 \mu\text{A}$

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Q15

Consider the LR circuit shown in the figure. If the switch S is closed at $t = 0$ then the amount of charge that passes through the battery between $t = 0$ and $t = \frac{L}{R}$ is :



- (1) $\frac{2.7EL}{R^2}$ (2) $\frac{EL}{2.7R^2}$ (3) $\frac{7.3EL}{R^2}$ (4) $\frac{EL}{7.3R^2}$

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EMI - Answers

Q1

[Bonus]

Net charge,

$$Q = \frac{\Delta\phi}{R} = \frac{1}{10} A(B_f - B_i) = \frac{1}{10} \times 3.5 \times 10^{-3} \left(0.4 \sin \frac{\pi}{2} - 0 \right)$$

$$= \frac{1}{10} (3.5 \times 10^{-3})(0.4 - 0)$$

$$= 1.4 \times 10^{-4}$$

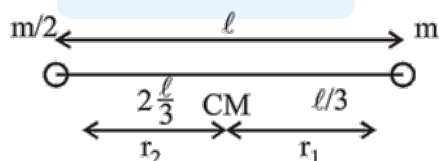
No option matches, so it should be a bonus.

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Q2

(3) As we know, $\omega = \sqrt{\frac{k}{I}}$

$$\omega = \sqrt{\frac{3k}{m\ell^2}} \quad \left[\because I_{\text{rod}} = \frac{1}{3} m\ell^2 \right]$$



Tension when it passes through the mean position,

$$= m\omega^2 \theta_0^2 \frac{\ell}{3} = m \frac{3k}{m\ell^2} \theta_0^2 \frac{\ell}{3} = \frac{k\theta_0^2}{\ell}$$

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Q3

(2) Efficiency, $\eta = \frac{P_{\text{out}}}{P_{\text{in}}} = \frac{V_s I_s}{V_p I_p}$

$$\Rightarrow 0.9 = \frac{230 \times I_s}{2300 \times 5}$$

$$\Rightarrow I_s = 0.9 \times 50 = 45 \text{ A}$$

Output current = 45 A

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Q4

(2) According to faraday's law of electromagnetic induction, $\epsilon = \frac{-d\phi}{dt}$

$$\epsilon = \frac{-d\phi}{dt}$$

$$L \times \frac{di}{dt} = 25 \Rightarrow L \times \frac{15}{1} = 25$$

$$\text{or, } L = \frac{5}{3} \text{ H}$$

Change in the energy of the inductance,

$$\Delta E = \frac{1}{2} L (i_1^2 - i_2^2) = \frac{1}{2} \times \frac{5}{3} \times (25^2 - 10^2)$$

$$= \frac{5}{6} \times 525 = 437.5 \text{ J}$$

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Q5

(4) The rate of mutual inductance is given by

$$M = \mu_0 n_1 n_2 \pi r_1^2 \quad \dots(i)$$

The rate of self inductance is given by

$$L = \mu_0 n_1^2 \pi r_1^2 \quad \dots(ii)$$

Dividing (i) by (ii)

$$\Rightarrow \frac{M}{L} = \frac{n_2}{n_1}$$

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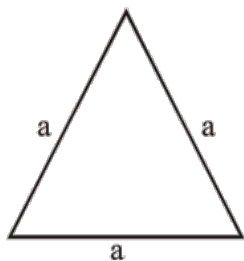
Q6

EMI

- (3) As total length L of the wire will remain constant

$$L = (3a) N \quad (N = \text{total turns})$$

and length of winding = $(d) N$



(d = diameter of wire)

self inductance = $\mu_0 n^2 A \ell$

$$= \mu_0 n^2 \left(\frac{\sqrt{3}a^2}{4} \right) dN$$

$$\propto a^2 N \propto a \quad [\text{as } N = L/3a \Rightarrow N \propto \frac{1}{a}]$$

Now ' a ' increased to ' $3a$ '

So self inductance will become 3 times

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Q7

- (1) Induced emf, $\varepsilon = Bv\ell$
 $= 0.3 \times 10^{-4} \times 5 \times 10$
 $= 1.5 \times 10^{-3} \text{ V}$

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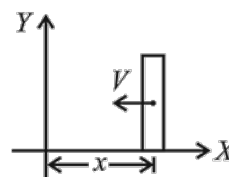
Q8

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- (3) Force on the strip when it is at stretched position x from mean position is

$$F = -kx - iIB = -kx - \frac{Biv}{R} \times IB$$

$$F = -kx - \frac{B^2 I^2}{R} \times v$$



Above expression shows that it is case of damped oscillation, so its amplitude can be given by

$$\Rightarrow A = A_0 e^{-\frac{bt}{2m}}$$

$$\Rightarrow \frac{A_0}{e} = A_0 e^{-\frac{bt}{2m}} \quad [\text{as per question } A = \frac{A_0}{e}]$$

$$\Rightarrow t = \frac{2m}{\left(\frac{B^2 I^2}{R} \right)} = \frac{2 \times 50 \times 10^{-3} \times 10}{0.01 \times 0.01}$$

$$= 10000 \text{ s}$$

Given, $m = 50 \times 10^{-3} \text{ kg}$

$$B = 0.1 \text{ T}$$

$$l = 0.1 \text{ m}$$

$$R = 10 \Omega$$

$$k = 0.5 \text{ N}$$

$$\text{Time period, } T = 2\pi \sqrt{\frac{m}{k}} \approx 2 \text{ s}$$

so, required number of oscillations,

$$N = \frac{10000}{2} = 5000$$

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Q9

$$(3) i^2 R = \left(\tau \frac{di}{dt} \right) i$$

$$\Rightarrow \frac{di}{dt} = \frac{i}{\tau}$$

$$\Rightarrow t = \tau \ln 2 = 2 \ln 2 \left[\text{as } \tau = \frac{L}{R} = \frac{20}{10} = 2 \right]$$

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Q10

$$(4) \text{ Inductance} = \frac{\mu_0 N^2 A}{L}$$

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Q11

$$\begin{aligned} (1) \quad B_0 &= \sqrt{B_0^2 + B_1^2} = \sqrt{30^2 + 2^2} \times 10^{-6} \\ &\approx 30 \times 10^{-6} \text{ T} \\ \therefore E_0 &= cB = 3 \times 10^8 \times 30 \times 10^{-6} \\ &= 9 \times 10^3 \text{ V/m} \\ E_{\text{rms}} &= \frac{E_0}{\sqrt{2}} = \frac{9}{\sqrt{2}} \times 10^3 \text{ V/m} \\ \text{Force on the charge,} \\ F &= E_{\text{rms}} Q = \frac{9}{\sqrt{2}} \times 10^3 \times 10^{-4} \approx 0.64 \text{ N} \end{aligned}$$

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Q11

$$\begin{aligned} (1) \quad Q &= BA \\ &= (\mu_0 n_i) A \\ &= \mu_0 n (kt e^{-\alpha t}) A \\ e &= -\frac{dQ}{dt} \\ &= -\mu_0 n A k \frac{d}{dt} (te^{-\alpha t}) \\ &= -\mu_0 n A k [t(-1)e^{-\alpha t} + e^{-\alpha t} \times 1] \\ &= -\mu_0 n A k [e^{-\alpha t} (1-t)] \\ i &= \frac{e}{R} = \frac{-\mu_0 n A k}{R} [e^{-\alpha t} (1-t)] \\ \text{At } t=0, i &\Rightarrow -ve \end{aligned}$$

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Q12

$$\begin{aligned} (1) \quad Q_{\text{coil}} &= (NQ) \propto i \\ \text{So, } \frac{Q_1}{Q_2} &= \frac{i_1}{i_2} = \frac{3}{2} \\ \text{or } Q_2 &= \frac{2}{3} Q_1 = \frac{2}{3} \times 10^{-3} = 6.67 \times 10^{-4} \text{ Wb} \end{aligned}$$

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Q13

$$(3) \text{ Power output } (V_2 I_2) = 2.2 \text{ kW}$$

$$\therefore V_2 = \frac{2.2 \text{ kW}}{(10 \text{ A})} = 220 \text{ volts}$$

$\therefore$ Input voltage for step-down transformer

$$\frac{V_1}{V_2} = \frac{N_1}{N_2} = 2$$

$$V_{\text{input}} = 2 \times V_{\text{output}} = 2 \times 220 = 440 \text{ V}$$

$$\text{Also } \frac{I_1}{I_2} = \frac{N_2}{N_1}$$

$$\therefore I_1 = \frac{1}{2} \times 10 = 5 \text{ A}$$

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Q14

$$(4) \quad I = I_0 \left(1 - e^{-\frac{Rt}{L}} \right) \quad \text{Here } R = R_L + r = 1 \Omega$$

$$0.8 I_0 = I_0 \left(1 - e^{-\frac{t}{.01}} \right)$$

$$\Rightarrow 0.8 = 1 - e^{-100t}$$

$$\Rightarrow e^{-100t} = 0.2 = \left(\frac{1}{5} \right)$$

$$\Rightarrow 100t = \ln 5 \Rightarrow t = \frac{1}{100} \ln 5 = 0.016 \text{ s}$$

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Q14

(2) Induced emf,

$$e = Bv\ell = 1 \times 10^{-2} \times 0.05 = 5 \times 10^{-4} \text{ V}$$

Equivalent resistance,

$$R = \frac{4 \times 2}{4 + 2} + 1.7$$

$$= \frac{4}{3} + 1.7 \approx 3 \Omega$$

$$\text{Current, } i = \frac{e}{R} = \frac{5 \times 10^{-4}}{3} \approx 170 \mu \text{ A}$$

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Q15

(2) We have, $i = i_0 (1 - e^{-t/\tau}) = \frac{\varepsilon}{R} (1 - e^{-t/\tau})$

Charge, $q = \int_0^{\tau} i dt$

$$= \frac{\varepsilon}{R} \int_0^{\tau} (1 - e^{-t/\tau}) dt = \frac{E \tau}{R e} = \frac{E}{R} \times \frac{(L/R)}{e}$$

$$= \frac{EL}{2.7 R^2}$$

12 April Evening



mathongo